

Formal Methods for Autonomous Vehicles

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This article describes the impact that autonomous vehicle operation will have on a range of domains and how formal methods can guarantee the safe operation of those vehicles. The verification of the safe operation of autonomous systems (e.g., autopilot systems for commercial aircraft) was traditionally done using precise models and verification software based on formal methods. However, the increasing use of machine-learning-based subsystems (e.g., for perception) requires the use of different verification tools. We describe how formal methods are used today and how they will be adapted for future autonomous vehicles.

Autonomous operation represents a significant restructuring of several sectors, such as automotive, commercial aviation, shipping, farming, etc. This constitutes a multitrillion dollar venture and will introduce major changes. A key change is to reduce transportation fatalities significantly. For example, in the automotive sector, it is estimated that autonomy can dramatically reduce crashes since 94% of serious crashes are due to human error.^a

This new autonomous vehicle (AV) sector represents the confluence of 1) traditional engineering and control theory with 2) AI, such that we can prove key properties using formal methods. As such, it is a microcosm of the change of modern cyberphysical systems (systems with a cyber and a physical component) being driven primarily from traditional engineering and control theory to AI- and data-driven software. What is different in the AV ecosystem from other systems (e.g., social networks, fintech and banking, etc.) is that a strong regulatory environment already exists to ensure safety. We focus on the use of formal methods to maintain the safety of passengers, pedestrians, vehicle operators, etc. Formal methods have traditionally been the primary method for ensuring safety, and this article

examines how safety verification is changing in this sector as autonomous operations make inroads into general system operation.

Ensuring safety for systems with subsystems based on machine learning (ML) and, in particular, deep learning (DL) is challenging. This is because ML/DL systems are often not explainable; e.g., they cannot be expressed as a transparent formal model like a temporal logic (TL). Further, their potential nondeterministic behavior makes the deployment in a safety-critical context troublesome.¹ While different approaches and frameworks have been proposed to overcome this issue,² the safety metrics employed in the ML/DL domain are still inadequate to provide well-defined vehicle-level safety requirements. The safety criteria used by ML-based subsystems must be verified using methods different from those employed by traditional formal methods techniques, and it is difficult to define well-founded, integrated, system-level metrics. However, given the prevalence of ML-based modules, it is critical to integrate available ML quality methods and metrics into the safety lifecycle and explicitly show their contribution to safety.²

Our contributions are as follows:

- ▶ We define autonomy and the role that formal methods play in verifying the behavior of AVs.
- ▶ We outline the new architectures that integrate traditional and ML-based modules.
- ▶ We highlight the paths toward full autonomy in some key sectors, namely, the automotive, commercial aviation, maritime, and farming sectors.

^a"Automated Vehicles for Safety, 2017." Available at <https://www.nhtsa.gov/technology-innovation/automated-vehicles-safety>.

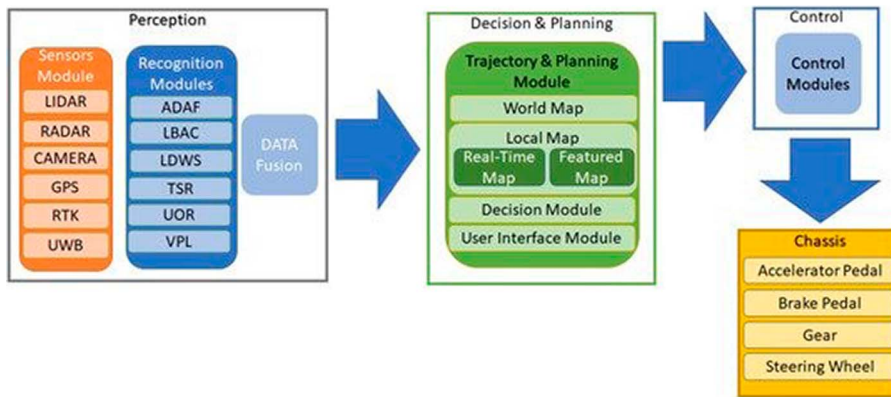


FIGURE 1. Generic system architecture for an autonomous car with blended capability.

WHAT IS AUTONOMY IN A FORMAL SENSE?

The section describes what autonomy is, the notion of safety, and how safe autonomy can be achieved.

Autonomy, roughly speaking, is the ability of a vehicle to execute tasks without human intervention. The tasks may vary by sector. For example, in commercial aviation, the task is to fly an aircraft with passengers from a starting gate at an airport to a gate at the destination airport. In factory scenarios, the task may be for an autonomous crane to locate goods in a warehouse, lift them, and deliver them to a truck at the loading bay.

An autonomous system will take sensor data as inputs and use these data to perform particular tasks. For example, an autonomous car has the task of safely driving human passengers from point A to point B, potentially with some engagement from a human—which we define as blended autonomy. Figure 1, adapted from Ahangar et al.,³ illustrates the overall architecture for an autonomous car with blended capability. It shows the sensors, actuators, hardware, and controllers required for vehicle operation; the chassis module can be used by a robot or human to control the vehicle, thus enabling blended capability.

Level of Autonomy

For the foreseeable future, it is expected that autonomy will exist in collaboration with human operators (blended autonomy). Within this context, we can define the degree of autonomy in terms of a level. For example, an autonomous system might be described as semiautonomous or fully autonomous. A semiautonomous robotic system is one where the “robot and a human operator plan and conduct the task, requiring various levels of human interaction.”⁴ A fully

autonomous robotic system performs its task “without human intervention, while adapting to operations and environmental conditions.”⁴ Different autonomy levels can have characteristic levels of safety; see, e.g., Clothier et al.⁵

Different application domains have defined their own set of levels. The automotive sector has defined six levels,⁶ the maritime sector has defined four levels,^b and the military has a range of levels.⁷

Formal Methods for Safety Guarantees

Formal methods are critical to the development of autonomous systems since they use software to make decisions *without human control*. Moreover, verification of the behavior of autonomous systems is especially important when they are embedded in safety-critical systems, which are increasingly being introduced into everyday settings.

Safety is a key goal of AVs. Here, a “safety specification” refers to a formal specification of the following form: “At all times, an undesired outcome never happens.”⁸ Safety is typically formalized as a specific kind of TL formula, e.g., linear TL, computation tree logic,⁹ or signal TL.¹⁰ For a recent survey, see Luckcuck.¹¹

Formalizing the rules of vehicle operation is a challenging task and is a key challenge for applying formal methods to AVs. Most rules of vehicle operation are originally stated in natural language, e.g., in traffic laws or driving manuals, which can differ among countries and localities within countries. Some rules of vehicle operation can be formalized as simple safety specifications that guarantee safety when satisfied.⁸ For example, a safety specification might be a formal rule that a vehicle should maintain a given minimum clearance

^b<https://www.imo.org/en/MediaCentre/HotTopics/Pages/Autonomous-shipping.aspx>

from pedestrians on the road at all times. Safety specifications are sometimes defined equivalently as reachability specifications. In the previous example, the reachability specification is that the vehicle must avoid distances to pedestrians that are smaller than the minimum clearance.

Computing Safety Guarantees

Autonomous system verification uses both gray-box models (combining statistics with formal rules, e.g., Fan et al.¹²) as well as traditional model-based approaches, e.g., Mehdi-pour et al.⁸ The model typically consists of a vehicle and its environment in autonomous driving. Verification entails checking the satisfaction of a model against all possible behaviors.

The literature at present uses safety specifications to perform formal verification and focuses on trajectory (or vehicle) following and lane keeping/changing.⁸ The formal methods used include 1) “traditional” methods, such as *theorem proving* and *model checking*; 2) *barrier certificates*; 3) *reachability analysis*; 4) *runtime verification*; and 5) *formal synthesis*.

Regulatory and Legal Issues

AVs that operate in regulated domains have to comply not only with technical requirements but also with requirements originating from regulatory and legal issues. These issues are domain specific, and in the “*Paths to Autonomy by Sector*” section, we examine several such issues for key sectors.

The key challenge, common with needing to precisely define rules of vehicle operation, is to translate the regulatory and legal issues into a common specification language with the technical and operational requirements.

ARCHITECTURES FOR AUTONOMOUS SYSTEMS

Due to the success of DL and other AI technologies, the means of achieving safety guarantees, both formally and operationally, is changing. This section examines how verification must change due to the introduction of ML/DL modules.

Verification of Architectures Integrating DL Modules

In a “traditional” control system, model-based controllers would be used to create the control module in Figure 1. These controllers can be rigorously analyzed using formal methods, and the performance of the control outputs is verified. In a system with ML/DL capability, the controller can be partially or totally replaced by

a learned representation, e.g., as generated via reinforcement learning.¹³ More commonly, the perception modules today consist of DL-based modules.

The verification challenges increase with mixed formal/ML/DL modules and with blended autonomy, where humans may interact and partially control a vehicle. Limited work exists in this area; for example, Bachute and Subhedar¹⁴ examine the task of integrating mixed modules. Kuutti et al.¹⁵ describe the strengths and limitations of available DL methods for AV control tasks; their work focuses on a comparative analysis of the research challenges using metrics such as computation, architecture selection, goal specification, generalization, verification/validation, and safety. Zhao et al.¹⁶ present a safety framework for DL that addresses quantitative claims based on structured heterogeneous safety arguments.

In this article, we focus on two key areas of AV operation, perception and control, both of which are currently being solved via DL.

- ▶ *AV control*: The standard DL-based method for AV control is reinforcement learning.^{13,17,18} Again, much work is necessary to map these methods to safety guarantees for AVs.
- ▶ *AV perception*: Correct perception of the environment is important, as incorrect outputs lead to unintended and often unsafe control actions.¹⁹ The inherent opacity of ML/DL processes and the perceived nondeterministic behavior make their deployment in a safety-critical context troublesome.¹ Recent work^{20,21,22} examines the use of formal methods for AV perception. For example, Schleiss et al.²³ address the task of specifying metrics for AV perception by providing a risk-based verification strategy, such as that required by ISO 21448.

Toward Certification of Mixed Systems

Given that all future AVs will consist of subsystems with mixed ML/DL and more traditional modules, it is critical to create mixed-system verification methods. Two key criteria are 1) metrics that define formal guarantees and 2) explainability.

- ▶ *Metrics*: Hanif et al.²⁴ present several techniques to make DL-based algorithms robust by analyzing and mitigating reliability and security threats. Borg et al.²⁵ review the state of the art in the verification and validation of safety-critical systems that rely on DL, with application to automotive software engineering.

- › **Explainability:** The concept of explainability concerns the ability of a human to understand the output of an ML/DL module. Any AV policy must identify and rely solely on true causes of expert decisions if we want it to be robust to a distributional shift between training and deployment situations. Explainability is even more important in safety-critical applications, like self-driving cars/aircraft.

PATHS TO AUTONOMY BY SECTOR

Different sectors are following quite different paths. This section provides a quick overview of some differing paths.

Automotive

Since the late 1990s, the technology sector has promised a world with driverless cars that efficiently transport people to their destinations. For example, Amnon Shashua, CEO of the autonomous driving technology company Mobileye, predicted in 2017 that no one would purchase automobiles.

These dreams have fueled billions of dollars in investment, but ubiquitous autonomous cars are still in development. The automotive sector has seen the aggressive development of autonomy, spearheaded both by vehicle manufacturers (like Tesla, GM, BMW, etc.) and by large technology companies (Waymo, Alphabet, etc.). These companies, together with

traditional automotive suppliers, all aim to provide level 5 vehicles.

The Society of Automotive Engineers has defined five levels of autonomy, ranging from level 1 (no autonomy) to level 5 (full autonomy)—see Figure 2. Currently, some vehicles (like Tesla) operate with level 4 autonomy, which enables driverless control under certain conditions.

However, setbacks have delayed the launch of level 4 and 5 AVs and delayed customer adoption. Nonetheless, the mobility community still thinks that autonomous driving has the potential to transform transportation, consumer behavior, and society in general.

To realize the consumer and commercial benefits of autonomous driving, auto OEMs and suppliers may need to develop new sales and business strategies, acquire new technological capabilities, and address concerns about safety. Safety standards have been modified as the automotive industry increases automation to levels 3, 4, and 5. The inclusion of ML, e.g., in the perception system, has prompted changes to safety standards. The functional safety standard ISO 26262²⁶ is a key element of automotive development. It deals with hazards caused by technical failures arising due to random and systematic faults in the system's hardware or software. ISO 26262 enables the computation of a safety goal with a discrete safety integrity level. However, the existing version does not fully cover the safety requirements of AV features that rely on environmental perception or ML/DL.²⁷ A major issue with ML/DL-based systems is that they are not fully

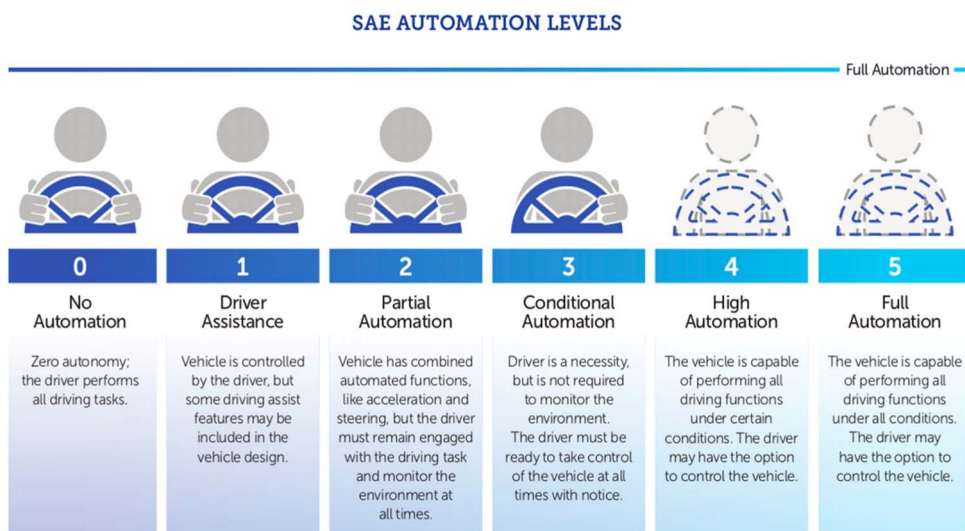


FIGURE 2. Society of Automotive Engineers levels of autonomy for the automotive domain. (Adapted from Ahangar et al.³; used with permission.)

specifiable, while ISO 26262 implicitly assumes that all functionality is specified.²⁸

In summary, due to the technical, legal, and other challenges just described, achieving level 5 autonomy remains at least five to 10 years away.

Commercial Aviation

Unlike the automotive sector, where there is a big push for full automation, commercial airlines are progressing very slowly.²⁹ This sector already has significant autonomy (in autopilot systems), and there are strong safety regulations already being enforced.

This sector is characterized by model-based certification of autonomy, where formal methods are used to guarantee operations of autonomous systems; see, e.g., the work by Cummings and Britton.³⁰ This is a time-consuming process and has relied on the use of two pilots onboard every flight. To reduce costs for pilots, air carriers have been developing plans to introduce single-pilot operations (SPO) as an option for the future evolution of commercial airliners. In other words, commercial airlines are focusing on moving from the current two-pilot operations to SPO prior to a consideration of full autonomy.³¹

In the SPO concept of operations, a single pilot operates the flight deck with increased ground support from a dedicated ground human flight crew. The ground operators fulfill a role similar to that of a remotely piloted aircraft system operator, providing a combination of strategic and tactical support to the single pilot in collaboration with the air traffic controllers.³¹

It is expected that this sector will proceed slowly, first to SPO and then to fully autonomous operations. This is primarily due to regulatory issues rather than technical obstacles.

Shipping

Shipping is progressing in a slow but steady manner toward increased automation. The maritime autonomous surface ships (MASS) sector views blended autonomy, rather than full automation, as the key to realizing intelligent navigation. This sector has defined four autonomy levels, with level 4 being fully autonomous navigation. Limited by the development of smart technology and maritime regulations, level 1–3 autonomous surface ships will coexist with humans and machines for a long time.³²

At present, blended autonomy for the navigation of MASS has been facing several challenges. These include blended autonomy control mode and risk assessment, the enhancement of blended autonomy situation awareness, decision making with blended

autonomy cooperative evolution, and providing safety guarantees switching between human navigation mode and machine navigation mode.³²

Hence, although significant research is underway to increase the level of autonomy, many obstacles arise due to the need for human–machine cooperation.³²

Farming

Of the sectors investigated, farming may be the most complex since it involves many types of vehicles (tractors, unmanned aerial vehicles, etc.), actors (humans and animals), and terrain.³³ For example, fully autonomous tractors work well in large, flat, rectangular fields but less well in smaller, hilly fields with complex geometry.

Overviews of the current technology readiness and of future plans can be found in Eaton et al.³⁴ and Bacco et al.³⁵ Ghobadpour et al.³⁶ present a review of trends, challenges, and opportunities in the use of AVs in the agricultural sector. Future trends indicate that autonomy will increase but at a slower pace compared to that of the automotive sector.

SUMMARY

The adoption of AVs with different levels of autonomy will continue to increase across all the sectors analyzed, leading to the increased safety of vehicle operators, passengers, pedestrians, and animals. A key driving factor is the introduction of better perception systems, typically driven by DL.

The need for precise and accurate perception systems increases drastically to ensure the safety of the passengers, pedestrians, and the surrounding vehicles' drivers. The future of autonomous perception relies on developing more robust algorithms trained on powerful computers and embedded on powerful "AI chips." Such devices would improve the ML models' efficiency, accuracy, and speed. The scalability of the existing methods is an important area of future investigation. This needs to be allied with more transparent methods for verifying the outputs of the black-box DL technology.

Overall, significant research is still necessary to enable safety certification of mixed-model-based and ML/DL systems. It is likely that the automotive sector will be the fastest to achieve the goal of full autonomy, and the technology will trickle down to sectors such as shipping and farming.

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